CSC311 Introduction to Machine Learning   Week 4 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W4-T1. Explain how we can use a linear regression model to learn a nonlinear function from an original input feature x. What does the word "linear" in linear regression mean?

W4-T2. Suppose you are training a regression model using a regularized cost function

$$\mathcal{E}_{\text{reg}}(\mathbf{w}) = \mathcal{E}(\mathbf{w}) + \lambda \mathcal{R}(\mathbf{w})$$

What could happen if λ is too large? What about too small?

W4-T3. During the lecture, we explored a few approaches for modeling a classification problem. Explain why each of the below models is problematic, and thus cannot be trained using gradient descent.

(a) Using linear regression and square loss, i.e.

$$\begin{aligned} z &= \mathbf{w}^T\mathbf{x} \\ y &= \mathbb{I}[z \geq 0] \\ \mathcal{L}_{SE} &= (z-t)^2 \end{aligned}$$

(b) Using a threshold activation function and 0-1 loss, i.e.

$$\begin{aligned} z &= \mathbf{w}^T\mathbf{x} \\ y &= \mathbb{I}[z \geq 0] \\ \mathcal{L}_{0-1} &= \mathbb{I}[y \neq t] \end{aligned}$$

Note: why is this loss equivalent $\mathcal{L} = (y-t)^2$?

(c) Using sigmoid activation function and square loss, i.e.

$$\begin{aligned} z &= \mathbf{w}^T\mathbf{x} \\ y &= \sigma(z) \\ \mathcal{L}_{SE} &= \frac{1}{2}(y-t)^2 \end{aligned}$$

W4-T4. What is the derivative of the sigmoid activation function? Show that $\sigma'(z) = \sigma(z)(1-\sigma(z))$

W4-T5. Compute $\nabla_{\mathbf{w}}\mathcal{L}_{CE}$ if we have the below model, by first computing $\frac{\partial \mathcal{L}}{\partial w_j}$ and vectorizing the result.

$$\begin{aligned} z &= \mathbf{w}^T\mathbf{x} \\ y &= \sigma(z) \\ \mathcal{L}_{CE}(y,t) &= -t\log(y) - (1-t)\log(1-y) \end{aligned}$$

W4-T6. Suppose that we have a logistic regression model $y = \sigma(\mathbf{w}^T\mathbf{x} + b)$ where $\mathbf{w} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ and $b = -0.5$. Draw the decision boundary for this model.

Validation

W4-V1. Consider a linear regression problem where we wish to fit $y = \mathbf{w}^T\mathbf{x}$. Where $\mathbf{x}$ is an input vector and $\mathbf{w}$ is the weight vector. As is typical, we use a training data set $\{(\mathbf{x}^{(1)}, t^{(1)}), (\mathbf{x}^{(2)}, t^{(2)}), \ldots, (\mathbf{x}^{(N)}, t^{(N)})\}$, and the MSE cost function $\mathcal{E}(\mathbf{w}) = \frac{1}{2N}\sum_i(\mathbf{w}^T\mathbf{x}^{(i)} - t^{(i)})^2$ computed on this data set. However, we would also like to use a L^2 regularizer $\mathcal{R}(\mathbf{w}) = \frac{1}{2}\|\mathbf{w}\|_2^2 = \frac{1}{2}\sum_j w_j^2$. Thus, we wish to find weights $\mathbf{w}$ that minimizes the regularized cost function across our training set:

$$\mathcal{E}_{\text{reg}}(\mathbf{w}) = \mathcal{E}(\mathbf{w}) + \lambda\mathcal{R}(\mathbf{w}).$$

(a) Showing all your work, show that

$$\frac{\partial\mathcal{E}_{\text{reg}}}{\partial w_j} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{w}^T\mathbf{x}^{(i)} - t^{(i)})x_j^{(i)} + \lambda w_j.$$

Do not need to repeat the vectorization work we did during our lectures.

(b) Use the above answer to write down the vectorized equation for the gradient $\nabla_{\mathbf{w}}\mathcal{E}_{\text{reg}}$. Use the notation $\mathbf{X}$ for the data matrix, $\mathbf{t}$ for the vector of targets, and $\mathbf{w}$ for the weight vector. (Aside: This type of regularization is also called **weigh decay**, and your answer to this question should explain why!)

(c) Find the direct solution to $\nabla_{\mathbf{w}}\mathcal{E}_{\text{reg}} = \mathbf{0}$ for the optimal $\mathbf{w}$.

(d) How would your answer from parts (b) and (c) change if, instead of having a single λ parameter, we had a different value of λ_j for each weight w_j? In this case, our revised cost function and gradients are:

$$\mathcal{E}_{\text{reg2}}(\mathbf{w}) = \mathcal{E}(\mathbf{w}) + \frac{1}{2}\sum_{j=1}^{D}\lambda_j w_j^2$$

$$\frac{\partial\mathcal{E}_{\text{reg2}}}{\partial w_j} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{w}^T\mathbf{x}^{(i)} - t^{(i)})x_j^{(i)} + \lambda_j w_j$$

Please the notation $\boldsymbol{\lambda} = \begin{bmatrix}\lambda_1 & \lambda_2 & \ldots & \lambda_D\end{bmatrix}^T$ for vectorization. You might also find the math prepare video on vectorization helpful for this question.

W4-V2. Consider the logistic regression model:

$$z = \mathbf{w}^T\mathbf{x}$$
$$y = \sigma(z)$$
$$\mathcal{L}_{CE}(y, t) = -t\log(y) - (1-t)\log(1-y)$$

We showed during lecture that if we were to minimize the average cross-entropy loss across our training data $\mathcal{E} = \frac{1}{N}\sum_{i=1}^{N}\mathcal{L}_{CE}(y^{(i)}, t^{(i)})$, then our gradient descent update rule will be as follows:

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{N}\sum_{i=1}^{N}(y^{(i)} - t^{(i)})\mathbf{x}^{(i)}$$

Determine the gradient descent update rule if we were to instead optimize the following regularized cost function

$$\mathcal{E}_{\text{reg}}(\mathbf{w}) = \mathcal{E}(\mathbf{w}) + \lambda_1||\mathbf{w}||_1 + \lambda_2||\mathbf{w}||_2$$

Where $||\mathbf{w}||_1 = \sum_{j=1}^{D}|w_j|$ is the L_1 norm $||\mathbf{w}||_2 = \sqrt{\sum_{j=1}^{D}w_j^2}$ is the L_2 norm, and λ_1, λ_2 are hyper-parameters.

Generalization

W4-G1. For this question, we would like to perform classification on the following training data:

$$\mathcal{D} = \{(x^{(1)} = 1, t^{(1)} = 0), (x^{(2)} = 2, t^{(2)} = 1)\}.$$

Euclidean distance will be used in all parts of this question.

(a) Draw the decision boundary of a 1NN model trained on this data set. Label the predictions in each region. (This is a review from previous weeks, but it's useful to set up the next part.)

(b) Suppose that we use a feature mapping function $\Psi(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix}$ to first map our training data $\mathcal{D}$ into a higher-dimensional space, and then perform 1NN on these new features. What is the new decision boundary, expressed in terms of the original data space $\mathbb{R}$?

You might find it helpful to recall that the solution to a quadratic equation of the form $ax^2 + bx + c = 0$ where $a \neq 0$ is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

W4-G2. Here we consider a medical test for a certain type of cancer. The test result is a single real number x. For people with the cancer, the value of x is normally distributed with a mean of a and a standard deviation of σ. For people without the cancer, the value of x is normally distributed with a mean of b and the same standard deviation σ. We will use a binary variable c to denote whether a person has cancer i.e., $c = 1$ means they have cancer, and $c = 0$ means they do not. Let p_1 be the proportion of the population that has the cancer. Thus, if we choose a person at random, the probability that they have the cancer is p_1.

(a) Write down expressions for the following quantities:

i. $P(c = 1)$

ii. $P(c = 0)$

iii. $P(x|c = 0)$

iv. $P(x|c = 1)$

v. $P(x, c = 0)$

vi. $P(x, c = 1)$

vii. $P(x)$

viii. $P(c = 1|x)$

(b) Given the test result x, show that the probability of having the cancer is logistic$(wx + d)$ for some constants w and d, where logistic$(z) = \frac{1}{(1+e^{-z})}$ is the logistic function (or sigmoid).

(c) Derive expressions for w and d.